Exercises on $D \sim \deg(D)[\infty]$ on $\mathbf{CP}^1$

computation, geometry, and divisor intuition

Setup

Let

$$X = \mathbf{CP}^1 = \mathbf{C} \cup \{\infty\}.$$

Use the affine coordinate z on $X \setminus \{\infty\}$, and near ∞ use

$$w = \frac{1}{z},$$

so that $w = 0$ corresponds to the point ∞.

For a nonzero meromorphic function f, its principal divisor is

$$(f) = \sum_{p \in X} \operatorname{ord}_p(f)[p].$$

Theorem 1. *For every divisor D on $\mathbf{CP}^1$,*

$$D \sim \deg(D)[\infty].$$

Equivalently, two divisors on $\mathbf{CP}^1$ are linearly equivalent if and only if they have the same degree.

Remark. *The point of the exercises is not just to know the theorem, but to feel why it is inevitable on $\mathbf{CP}^1$: every finite point can be moved to ∞ by a rational function, and all divisor-class information collapses to degree.*

Part I. Core computations

Exercise 1. *Compute the divisor of z on $\mathbf{CP}^1$. Show that*

$$(z) = [0] - [\infty].$$

Exercise 2. *For $a \in \mathbf{C}$, compute the divisor of $z - a$. Show that*

$$(z - a) = [a] - [\infty].$$

Exercise 3. *For distinct $a, b \in \mathbf{C}$, compute*

$$\left(\frac{z-a}{z-b}\right).$$

Deduce that

$$[a] \sim [b].$$

Exercise 4. *Compute the divisor of z^n for $n \in \mathbf{Z}$.*

Exercise 5. *Let*

$$f(z) = \prod_{j=1}^{r} (z - a_j)^{m_j},$$

where the a_j are distinct points of $\mathbf{C}$ and $m_j \in \mathbf{Z}$. Compute (f) explicitly.

Part II. Order at ∞

Exercise 6. *Let $P(z)$ be a nonzero polynomial of degree d. Show that*

$$\mathrm{ord}_\infty(P) = -d.$$

Exercise 7. *Let*

$$f(z) = \frac{P(z)}{Q(z)}$$

with nonzero polynomials P, Q. Show that

$$\mathrm{ord}_\infty(f) = \deg(Q) - \deg(P).$$

Exercise 8. *Prove directly that every principal divisor on $\mathbf{CP}^1$ has degree 0.*

Exercise 9. *Compute the divisor of*

$$f(z) = \frac{z^2+1}{z(z-1)^3}.$$

You must include the order at ∞.

Part III. Building $D \sim \deg(D)[\infty]$

Exercise 10. *Using $z - a$, prove that for each $a \in \mathbf{C}$,*

$$[a] \sim [\infty].$$

Exercise 11. *Let*

$$D = \sum_{j=1}^{r} n_j [a_j]$$

with $a_j \in \mathbf{C}$. Show directly that

$$D \sim \left(\sum_{j=1}^{r} n_j \right) [\infty].$$

Construct an explicit rational function whose divisor is

$$D - \deg(D)[\infty].$$

Exercise 12. *Now let*

$$D = \sum_{j=1}^{r} n_j [a_j] + m[\infty].$$

Show that

$$D \sim \deg(D)[\infty].$$

Exercise 13. *Prove that for divisors D, E on $\mathbf{CP}^1$,*

$$D \sim E \quad \Longleftrightarrow \quad \deg(D) = \deg(E).$$

Part IV. Concrete divisor engineering

Exercise 14. *Show that*

$$[0] + [1] + [2] - 3[\infty]$$

is principal, and construct an explicit rational function with this divisor.

Exercise 15. *Find an explicit rational function f such that*

$$(f) = 2[0] - [3] - [\infty].$$

Exercise 16. *Determine whether*

$$D = 3[1] - 2[i] + 5[-2] - 6[\infty]$$

is principal. If it is, construct f explicitly.

Exercise 17. *Let*

$$D = 2[0] - [1] + 4[\infty].$$

Find $n \in \mathbf{Z}$ such that

$$D \sim n[\infty],$$

and explicitly write a principal divisor equal to $D - n[\infty]$.

Exercise 18. *Give a necessary and sufficient condition for*

$$[p_1] + \cdots + [p_r] \sim [q_1] + \cdots + [q_s]$$

on $\mathbf{CP}^1$.

Part V. Through the spaces $L(D)$

For a divisor D, define

$$L(D) = \{f \in \mathcal{M}(X)^\times : (f) + D \geq 0\} \cup \{0\}.$$

Exercise 19. *Compute $L(n[\infty])$ for $n \geq 0$, and show that*

$$L(n[\infty]) = \operatorname{span}_{\mathbf{C}}\{1, z, z^2, \ldots, z^n\}.$$

Exercise 20. *Show that if $D \sim n[\infty]$ and $n \geq 0$, then*

$$\dim_{\mathbf{C}} L(D) = n + 1.$$

Exercise 21. *Compute $\dim_{\mathbf{C}} L(D)$ for:*

$$D = [0] + [1], \qquad D = [0] - [\infty], \qquad D = 2[\infty], \qquad D = -[\infty].$$

Exercise 22. *Let*

$$D = [a] + [b] + [\infty], \qquad a \neq b.$$

Show that $D \sim 3[\infty]$, and then find a basis of $L(D)$.

Part VI. Conceptual questions

Exercise 23. *Explain geometrically why ∞ can "collect" all zeros and poles on $\mathbf{CP}^1$.*

Exercise 24. *A student says: "If $D \sim \deg(D)[\infty]$, then all points of $\mathbf{CP}^1$ are literally the same." Write a careful correction distinguishing:*

- *equality of points,*
- *equality of divisors,*
- *linear equivalence of divisors.*

Exercise 25. *Explain why the statement*

$$D \sim \deg(D)[\infty]$$

fails on a general compact Riemann surface of genus $g > 0$.

Selected solutions

Selected solution (Exercise 1)**.** *The function z has a simple zero at 0, so*

$$\operatorname{ord}_0(z) = 1.$$

It has no zero or pole at any finite $a \neq 0$, so

$$\operatorname{ord}_a(z) = 0 \qquad (a \neq 0, \infty).$$

At ∞, use $w = 1/z$. Then

$$z = \frac{1}{w},$$

which has a simple pole at $w = 0$. Hence

$$\operatorname{ord}_\infty(z) = -1.$$

Therefore

$$(z) = [0] - [\infty].$$

Selected solution (Exercise 2)**.** *At $z = a$, the function $z - a$ vanishes simply, so*

$$\operatorname{ord}_a(z - a) = 1.$$

At any other finite point, the order is 0.

Near ∞, write

$$z - a = \frac{1 - aw}{w}.$$

Since $1 - aw$ is nonvanishing at $w = 0$, this has a simple pole at $w = 0$. Thus

$$\operatorname{ord}_\infty(z - a) = -1.$$

Hence

$$(z - a) = [a] - [\infty].$$

Selected solution (Exercise 3)**.** *Using the previous computation,*

$$\left(\frac{z-a}{z-b}\right) = (z-a) - (z-b) = ([a]-[\infty]) - ([b]-[\infty]) = [a]-[b].$$

So $[a]-[b]$ *is principal, which means*

$$[a] \sim [b].$$

Selected solution (Exercise 7)**.** *Write*

$$P(z) = a_m z^m + \cdots, \qquad Q(z) = b_n z^n + \cdots$$

with $a_m, b_n \neq 0$*, where* $m = \deg(P)$ *and* $n = \deg(Q)$*. Then near* ∞*,*

$$P(z) = w^{-m}(a_m + \cdots), \qquad Q(z) = w^{-n}(b_n + \cdots),$$

so

$$\frac{P(z)}{Q(z)} = w^{n-m} \cdot \frac{a_m + \cdots}{b_n + \cdots}.$$

The second factor is holomorphic and nonvanishing at $w = 0$*, hence*

$$\operatorname{ord}_\infty\left(\frac{P}{Q}\right) = n - m = \deg(Q) - \deg(P).$$

Selected solution (Exercise 11)**.** *Take*

$$f(z) = \prod_{j=1}^{r} (z - a_j)^{n_j}.$$

If some $n_j < 0$*, that factor contributes a pole instead of a zero, which is exactly what is needed.*

From the divisor formula for powers and products,

$$(f) = \sum_{j=1}^{r} n_j [a_j] - \left(\sum_{j=1}^{r} n_j\right)[\infty].$$

But

$$\deg(D) = \sum_{j=1}^{r} n_j,$$

so

$$(f) = D - \deg(D)[\infty].$$

Hence

$$D \sim \deg(D)[\infty].$$

Selected solution (Exercise 13)**.** *If* $D \sim E$*, then* $D - E$ *is principal, and every principal divisor has degree* 0*. Therefore*

$$\deg(D) = \deg(E).$$

Conversely, assume $\deg(D) = \deg(E)$*. Then*

$$\deg(D - E) = 0.$$

By the previous exercise,

$$D - E \sim 0,$$

since every degree-zero divisor on $\mathbf{CP}^1$ *is principal. Therefore*

$$D \sim E.$$

So on $\mathbf{CP}^1$*, linear equivalence classes are exactly degree classes.*

Selected solution (Exercise 14)**.** *Take*

$$f(z) = z(z-1)(z-2).$$

Then

$$(f) = (z) + (z-1) + (z-2)$$

and each term equals $[a] - [\infty]$ *for the appropriate* a*. Hence*

$$(f) = [0] + [1] + [2] - 3[\infty].$$

So the divisor is principal.

Selected solution (Exercise 15)**.** *We want zeros and poles matching*

$$2[0] - [3] - [\infty].$$

A double zero at 0 *comes from* z^2*, and a simple pole at* 3 *comes from dividing by* $z - 3$*. So try*

$$f(z) = \frac{z^2}{z-3}.$$

Then

$$(f) = 2[0] - [3] - [\infty],$$

because the numerator has degree 2 *and denominator degree* 1*, so* f *has a simple pole at* ∞*.*

Selected solution (Exercise 16)**.** *Its degree is*

$$3 - 2 + 5 - 6 = 0.$$

On $\mathbf{CP}^1$*, every degree-zero divisor is principal. An explicit function is*

$$f(z) = \frac{(z-1)^3(z+2)^5}{(z-i)^2 z^6}.$$

Indeed,

$$(f) = 3[1] + 5[-2] - 2[i] - 6[0].$$

This is not yet the divisor D*, because* D *has* $-6[\infty]$*, not* $-6[0]$*. So instead build it using the basic blocks* $z - a$*:*

$$f(z) = \frac{(z-1)^3(z+2)^5}{(z-i)^2}.$$

Then

$$(f) = 3[1] - 2[i] + 5[-2] - 6[\infty].$$

So D *is principal.*

Selected solution (Exercise 17)**.** *First compute the degree:*

$$\deg(D) = 2 - 1 + 4 = 5.$$

So

$$D \sim 5[\infty].$$

Now

$$D - 5[\infty] = 2[0] - [1] - [\infty].$$

An explicit function with this divisor is

$$f(z) = \frac{z^2}{z-1}.$$

Indeed,

$$(f) = 2[0] - [1] - [\infty].$$

Selected solution (Exercise 19)**.** *A rational function lies in* $L(n[\infty])$ *exactly when it has no poles except possibly at* ∞*, and its pole order there is at most* n*. Such functions are exactly polynomials of degree at most* n*. Therefore*

$$L(n[\infty]) = \{a_0 + a_1 z + \cdots + a_n z^n : a_j \in \mathbf{C}\},$$

so

$$L(n[\infty]) = \mathrm{span}_{\mathbf{C}}\{1, z, \ldots, z^n\}.$$

Selected solution (Exercise 22)**.** *The special feature of* $\mathbf{CP}^1$ *is that every point* $a \in \mathbf{C}$ *satisfies*

$$[a] - [\infty] = (z - a),$$

so each finite point can be transferred to ∞ *by a principal divisor. Repeating this for all points appearing in a divisor collapses the divisor to one supported only at* ∞*, and the only invariant left is total degree.*

On a higher-genus compact Riemann surface, degree-zero divisors need not be principal. So even if $\deg(D) = 0$*, one cannot generally conclude* $D \sim 0$*. That is precisely why the statement*

$$D \sim \deg(D)[\infty]$$

is special to genus 0*.*

Very short summary to remember

On $\mathbf{CP}^1$,

$$(z - a) = [a] - [\infty].$$

That single computation implies:

$$[a] \sim [\infty] \quad \text{for every } a \in \mathbf{C},$$

hence every divisor is equivalent to one supported at ∞, namely

$$D \sim \deg(D)[\infty].$$

So

$$\mathrm{Div}(\mathbf{CP}^1)/\sim \;\cong\; \mathbf{Z},$$

with isomorphism given by degree.

Winding-form proof of $D \sim \deg(D)[\infty]$ on $\mathbf{CP}^1$. For a meromorphic function f and a small positively oriented loop γ_p around p,

$$\operatorname{ord}_p(f) = \frac{1}{2\pi i} \int_{\gamma_p} \frac{df}{f}.$$

Thus the principal divisor of f records the local winding numbers of f around 0:

$$(f) = \sum_{p \in \mathbf{CP}^1} \left(\frac{1}{2\pi i} \int_{\gamma_p} \frac{df}{f} \right) [p].$$

For $a \in \mathbf{C}$, take $f(z) = z - a$. Around a,

$$\frac{1}{2\pi i} \int_{\gamma_a} \frac{d(z-a)}{z-a} = 1,$$

while near ∞, using $w = 1/z$,

$$z - a = \frac{1 - aw}{w}, \qquad \frac{d(z-a)}{z-a} = d\log(1-aw) - d\log w,$$

so

$$\frac{1}{2\pi i} \int_{\gamma_\infty} \frac{d(z-a)}{z-a} = -1.$$

Hence

$$(z-a) = [a] - [\infty],$$

so $[a] \sim [\infty]$.

Now let

$$D = \sum_j n_j [a_j].$$

Set

$$f(z) = \prod_j (z - a_j)^{n_j}.$$

Then

$$\frac{df}{f} = \sum_j n_j \frac{d(z-a_j)}{z-a_j}.$$

Therefore

$$\operatorname{ord}_{a_j}(f) = n_j, \qquad \operatorname{ord}_\infty(f) = -\sum_j n_j.$$

So

$$(f) = \sum_j n_j [a_j] - \left(\sum_j n_j \right) [\infty] = D - \deg(D)[\infty].$$

Thus

$$D \sim \deg(D)[\infty].$$

Exercises on $D \sim \deg(D)[\infty]$ via the winding form on $\mathbf{CP}^1$

Guiding principle

Let

$$X = \mathbf{CP}^1 = \mathbf{C} \cup \{\infty\}.$$

For a nonzero meromorphic function f on X, and a small positively oriented loop γ_p around a point $p \in X$, we have

$$\operatorname{ord}_p(f) = \frac{1}{2\pi i} \int_{\gamma_p} \frac{df}{f}.$$

So the divisor

$$(f) = \sum_{p \in X} \operatorname{ord}_p(f)[p]$$

can be read entirely from integrals of the winding form

$$\frac{df}{f}.$$

The goal of these exercises is to understand the statement

$$D \sim \deg(D)[\infty]$$

by interpreting divisors as local winding-number data.

Coordinates and conventions

Use the affine coordinate z on $X \setminus \{\infty\}$, and near ∞ use

$$w = \frac{1}{z},$$

so that $w = 0$ corresponds to the point ∞.

For each point p, let γ_p denote a sufficiently small positively oriented loop around p.

Part I. Local winding numbers as orders

1. Let $f(z) = z$. Compute

$$\frac{1}{2\pi i} \int_{\gamma_0} \frac{dz}{z}.$$

Then compute the corresponding integral around ∞ by rewriting z in the coordinate $w = 1/z$. Conclude that

$$(z) = [0] - [\infty].$$

2. Let $f(z) = z - a$ for $a \in \mathbf{C}$. Compute

$$\frac{1}{2\pi i} \int_{\gamma_a} \frac{d(z-a)}{z-a}.$$

Then pass to the coordinate $w = 1/z$ and compute

$$\frac{1}{2\pi i} \int_{\gamma_\infty} \frac{d(z-a)}{z-a}.$$

Conclude that

$$(z - a) = [a] - [\infty].$$

3. Let
$$f(z) = \frac{z-a}{z-b}, \qquad a \neq b.$$
Compute
$$\frac{df}{f} = \frac{d(z-a)}{z-a} - \frac{d(z-b)}{z-b}.$$
Use loop integrals to determine the order of f at a, b, and ∞. Deduce that
$$\left(\frac{z-a}{z-b}\right) = [a] - [b].$$

4. Let $m \in \mathbf{Z}$ and $f(z) = (z-a)^m$. Show directly from
$$\frac{df}{f} = m\frac{d(z-a)}{z-a}$$
that
$$\mathrm{ord}_a(f) = m.$$

5. Suppose near p one has
$$f = (\zeta - \zeta(p))^m u$$
in a local coordinate ζ, where u is holomorphic and nowhere zero. Show that
$$\frac{df}{f} = m\frac{d\zeta}{\zeta - \zeta(p)} + \frac{du}{u},$$
and explain why
$$\frac{1}{2\pi i}\int_{\gamma_p} \frac{df}{f} = m.$$
This is the local argument principle in divisor language.

Part II. Understanding ∞ through the winding form

6. Rewrite
$$z - a = \frac{1-aw}{w}$$
in the coordinate $w = 1/z$, and compute
$$\frac{d(z-a)}{z-a} = d\log(1-aw) - d\log w.$$
Use this to explain why the winding contribution at ∞ is -1.

7. Let $P(z)$ be a polynomial of degree d. Show, using $w = 1/z$, that
$$\mathrm{ord}_\infty(P) = \frac{1}{2\pi i}\int_{\gamma_\infty} \frac{dP}{P} = -d.$$

8. Let
$$f(z) = \frac{P(z)}{Q(z)}$$
for nonzero polynomials P, Q. Show that
$$\frac{df}{f} = \frac{dP}{P} - \frac{dQ}{Q},$$
and deduce from winding integrals that
$$\mathrm{ord}_\infty(f) = \deg(Q) - \deg(P).$$

9. Let
$$f(z) = \frac{(z-a_1)\cdots(z-a_r)}{(z-b_1)\cdots(z-b_s)}.$$
Compute $\mathrm{ord}_\infty(f)$ from the winding form. Explain why:
   - f has a zero at ∞ if $s > r$,
   - f has a pole at ∞ if $r > s$,
   - f is nonvanishing and finite at ∞ if $r = s$.

10. Compute the divisor of
$$f(z) = \frac{z^2+1}{z(z-1)^3}$$
entirely by integrating $\frac{df}{f}$ around the relevant points.

Part III. Global balance: why degree appears

11. Let f be a nonzero meromorphic function on $\mathbf{CP}^1$. Explain why the residues of $\frac{df}{f}$ are exactly the integers $\mathrm{ord}_p(f)$.

12. Use the residue theorem on $\mathbf{CP}^1$ to prove
$$\sum_{p\in\mathbf{CP}^1} \mathrm{ord}_p(f) = 0.$$
Interpret this as the statement
$$\deg(f) = 0.$$

13. Explain in words why the total positive winding and total negative winding must balance on a compact Riemann surface.

14. Show that if
$$(f) = \sum_p n_p[p],$$
then
$$\sum_p n_p = 0.$$
Why is this the precise reason that principal divisors preserve degree?

15. Let
$$f(z) = \prod_{j=1}^{r} (z-a_j)^{n_j}.$$
Compute
$$\frac{df}{f} = \sum_{j=1}^{r} n_j \frac{d(z-a_j)}{z-a_j}.$$
Use winding integrals to show
$$(f) = \sum_{j=1}^{r} n_j[a_j] - \left(\sum_{j=1}^{r} n_j\right)[\infty].$$

Part IV. From winding data to $D \sim \deg(D)[\infty]$

16. Using the function $z - a$, explain through winding integrals why

$$[a] \sim [\infty].$$

17. Let

$$D = \sum_{j=1}^{r} n_j [a_j]$$

with $a_j \in \mathbf{C}$. Consider

$$f(z) = \prod_{j=1}^{r} (z - a_j)^{n_j}.$$

Show, by computing local winding numbers, that

$$(f) = D - \deg(D)[\infty].$$

Conclude that

$$D \sim \deg(D)[\infty].$$

18. Now let

$$D = \sum_{j=1}^{r} n_j [a_j] + m[\infty].$$

Use the previous exercise to show carefully that

$$D \sim \deg(D)[\infty].$$

19. Prove that on $\mathbf{CP}^1$, two divisors D, E are linearly equivalent if and only if

$$\deg(D) = \deg(E).$$

Your proof must explicitly use the integral interpretation of orders.

20. Explain why the theorem says that on $\mathbf{CP}^1$, a divisor class is exactly the same thing as the total net winding stored at ∞.

Part V. Concrete winding-engineering problems

21. Construct a rational function f whose winding form satisfies

$$\frac{1}{2\pi i} \int_{\gamma_0} \frac{df}{f} = 1, \qquad \frac{1}{2\pi i} \int_{\gamma_1} \frac{df}{f} = 1, \qquad \frac{1}{2\pi i} \int_{\gamma_2} \frac{df}{f} = 1.$$

What must the winding at ∞ be? Write the full divisor of f.

22. Construct f such that

$$\frac{1}{2\pi i} \int_{\gamma_0} \frac{df}{f} = 2, \qquad \frac{1}{2\pi i} \int_{\gamma_3} \frac{df}{f} = -1.$$

Determine automatically what the winding at ∞ must be, and write (f).

23. Let
$$D = 3[1] - 2[i] + 5[-2] - 6[\infty].$$
Construct f with $(f) = D$ by matching the required local winding numbers.

24. Let
$$D = 2[0] - [1] + 4[\infty].$$
Find n such that
$$D \sim n[\infty].$$
Then explicitly construct a function f whose winding data realizes
$$D - n[\infty].$$

25. Give a necessary and sufficient condition, in terms of total winding number, for
$$[p_1] + \cdots + [p_r] \sim [q_1] + \cdots + [q_s]$$
on $\mathbf{CP}^1$.

Part VI. Conceptual exercises

26. A principal divisor is the collection of local winding numbers of a global meromorphic function. Explain this statement carefully.

27. Why does the function $z - a$ show that a single unit of winding can be moved from a to ∞?

28. Explain why ∞ acts as the unique "balancing point" for winding data on $\mathbf{CP}^1$.

29. A student says: "Since $[a] \sim [\infty]$, the points a and ∞ are equal." Correct this using the difference between:
    - points,
    - divisors,
    - principal divisors,
    - winding data of meromorphic functions.

30. Explain why the same argument does *not* show that every degree-zero divisor is principal on a compact Riemann surface of genus $g > 0$. What special global feature of $\mathbf{CP}^1$ is being used?

Selected solutions

Solution to Exercise 2. Let $f(z) = z - a$. Around a,

$$\frac{1}{2\pi i}\int_{\gamma_a} \frac{d(z-a)}{z-a} = 1,$$

since $z - a$ winds once around 0.

At ∞, write

$$z - a = \frac{1 - aw}{w}.$$

Then

$$\frac{d(z-a)}{z-a} = d\log(1-aw) - d\log w.$$

The first term is holomorphic near $w = 0$, so its integral around γ_∞ is 0. The second contributes $-2\pi i$. Therefore

$$\frac{1}{2\pi i}\int_{\gamma_\infty} \frac{d(z-a)}{z-a} = -1.$$

Hence

$$(z-a) = [a] - [\infty].$$

Solution to Exercise 12. The meromorphic 1-form $\dfrac{df}{f}$ has residue

$$\operatorname{res}_p\left(\frac{df}{f}\right) = \operatorname{ord}_p(f)$$

at each point p. Since $\mathbf{CP}^1$ is compact, the residue theorem gives

$$\sum_{p\in\mathbf{CP}^1} \operatorname{res}_p\left(\frac{df}{f}\right) = 0.$$

Therefore

$$\sum_{p\in\mathbf{CP}^1} \operatorname{ord}_p(f) = 0,$$

which is exactly

$$\deg(f) = 0.$$

Solution to Exercise 17. Let

$$D = \sum_{j=1}^{r} n_j[a_j]$$

and define

$$f(z) = \prod_{j=1}^{r} (z - a_j)^{n_j}.$$

Then

$$\frac{df}{f} = \sum_{j=1}^{r} n_j \frac{d(z-a_j)}{z-a_j}.$$

So for each a_k,

$$\frac{1}{2\pi i}\int_{\gamma_{a_k}} \frac{df}{f} = n_k.$$

Thus $\operatorname{ord}_{a_k}(f) = n_k$.

By global balance,

$$\operatorname{ord}_\infty(f) = -\sum_{j=1}^{r} n_j.$$

Hence

$$(f) = \sum_{j=1}^{r} n_j[a_j] - \left(\sum_{j=1}^{r} n_j\right)[\infty] = D - \deg(D)[\infty].$$

Therefore

$$D \sim \deg(D)[\infty].$$

Solution to Exercise 22. We want

$$(f) = 3[1] - 2[i] + 5[-2] - 6[\infty].$$

Take

$$f(z) = \frac{(z-1)^3(z+2)^5}{(z-i)^2}.$$

Then

$$\frac{df}{f} = 3\frac{d(z-1)}{z-1} + 5\frac{d(z+2)}{z+2} - 2\frac{d(z-i)}{z-i}.$$

So the local winding numbers are 3 at 1, 5 at -2, -2 at i, and the remaining balance at ∞ is

$$-(3+5-2) = -6.$$

Thus indeed

$$(f) = 3[1] - 2[i] + 5[-2] - 6[\infty].$$

One-line slogan

On $\mathbf{CP}^1$, a divisor is a prescribed pattern of local winding numbers, and a rational function can realize that pattern everywhere except for one necessary balancing term at ∞. That balancing term is exactly $\deg(D)[\infty]$.

Parallel handout: divisor language vs. winding-integral language on $\mathbf{CP}^1$

Purpose

This handout presents the statement

$$D \sim \deg(D)[\infty] \qquad \text{on } \mathbf{CP}^1$$

through two parallel viewpoints:

$$\textbf{divisor / order viewpoint} \iff \textbf{winding / integral viewpoint}$$

The point is that these are not two different theories. They are the same theory written in two different languages.

Setup

Let

$$X = \mathbf{CP}^1 = \mathbf{C} \cup \{\infty\}.$$

Use the affine coordinate z on $X \setminus \{\infty\}$, and near ∞ use

$$w = \frac{1}{z},$$

so that $w = 0$ corresponds to ∞.

For a nonzero meromorphic function f, define its principal divisor by

$$(f) = \sum_{p \in X} \operatorname{ord}_p(f)[p].$$

For a small positively oriented loop γ_p around p, one has

$$\operatorname{ord}_p(f) = \frac{1}{2\pi i} \int_{\gamma_p} \frac{df}{f}.$$

Dictionary

Divisor / order language	**Winding / integral language**
$\mathrm{ord}_p(f)$ is the zero/pole order of f at p.	$\mathrm{ord}_p(f) = \dfrac{1}{2\pi i}\displaystyle\int_{\gamma_p}\dfrac{df}{f}$ is the winding number of $f(\gamma_p)$ around 0.
A zero of order m contributes $+m[p]$.	A local winding number $+m$ contributes $+m$.
A pole of order m contributes $-m[p]$.	A local winding number $-m$ contributes $-m$.
(f) is the principal divisor of f.	(f) is the collection of all local winding numbers of f.
$\deg(f) = 0$.	$\displaystyle\sum_{p\in X}\dfrac{1}{2\pi i}\int_{\gamma_p}\dfrac{df}{f} = 0$, i.e. total winding balances globally.
$D \sim E$ means $D - E = (f)$.	$D \sim E$ means the difference between D and E is realized by winding data of a global meromorphic function.
$[a] - [\infty] = (z - a)$.	$z - a$ has winding $+1$ at a and winding -1 at ∞.

The basic model computation

Left column: divisor computation

Let $f(z) = z - a$. Then:

$$\mathrm{ord}_a(z - a) = 1.$$

Near ∞, using $w = 1/z$,

$$z - a = \frac{1 - aw}{w},$$

so $z - a$ has a simple pole at $w = 0$, hence

$$\mathrm{ord}_\infty(z - a) = -1.$$

Therefore

$$(z - a) = [a] - [\infty].$$

Right column: winding computation

Again let $f(z) = z - a$. Around a small loop γ_a,

$$\frac{1}{2\pi i}\int_{\gamma_a}\frac{d(z - a)}{z - a} = 1.$$

So f winds once around 0 near a.

Near ∞, write

$$z - a = \frac{1 - aw}{w}.$$

Then

$$\frac{d(z-a)}{z-a} = d\log(1-aw) - d\log w.$$

The first term contributes 0 around $w = 0$, while the second contributes $-2\pi i$. So

$$\frac{1}{2\pi i}\int_{\gamma_\infty} \frac{d(z-a)}{z-a} = -1.$$

Hence $z - a$ contributes one unit of winding at a and minus one unit at ∞.

$$\boxed{(z-a) = [a] - [\infty]}$$

This is the single most important computation in the entire story.

What $[a] \sim [\infty]$ means in the two languages

Divisor / order language	**Winding / integral language**
$[a] \sim [\infty]$ because $[a] - [\infty] = (z-a)$ is principal.	A single positive winding at a can be transferred to ∞ by the function $z - a$.

So linear equivalence here means: one can redistribute local winding data by multiplying by a global meromorphic function.

The ratio $\frac{z-a}{z-b}$

Left column

$$\left(\frac{z-a}{z-b}\right) = (z-a) - (z-b) = ([a] - [\infty]) - ([b] - [\infty]) = [a] - [b].$$

Hence

$$[a] \sim [b].$$

Right column

Let

$$f(z) = \frac{z-a}{z-b}.$$

Then

$$\frac{df}{f} = \frac{d(z-a)}{z-a} - \frac{d(z-b)}{z-b}.$$

So:

$$\frac{1}{2\pi i}\int_{\gamma_a} \frac{df}{f} = 1, \qquad \frac{1}{2\pi i}\int_{\gamma_b} \frac{df}{f} = -1,$$

and the contribution at ∞ cancels:

$$\frac{1}{2\pi i}\int_{\gamma_\infty} \frac{df}{f} = 0.$$

Thus the function creates one positive winding at a, one negative winding at b, and nothing else.

$$\boxed{\left(\frac{z-a}{z-b}\right) = [a] - [b]}$$

General divisor supported on finite points

Let

$$D = \sum_{j=1}^{r} n_j [a_j], \qquad a_j \in \mathbf{C}.$$

Define

$$f(z) = \prod_{j=1}^{r} (z - a_j)^{n_j}.$$

Left column

Using $(z - a_j) = [a_j] - [\infty]$,

$$(f) = \sum_{j=1}^{r} n_j ([a_j] - [\infty]) = \sum_{j=1}^{r} n_j [a_j] - \left(\sum_{j=1}^{r} n_j\right) [\infty].$$

Therefore

$$(f) = D - \deg(D)[\infty].$$

So

$$D \sim \deg(D)[\infty].$$

Right column

Since

$$\frac{df}{f} = \sum_{j=1}^{r} n_j \frac{d(z - a_j)}{z - a_j},$$

the local winding at a_j is n_j:

$$\frac{1}{2\pi i} \int_{\gamma_{a_j}} \frac{df}{f} = n_j.$$

The total winding over all points must sum to zero, so the contribution at ∞ is

$$-\sum_{j=1}^{r} n_j = -\deg(D).$$

Hence the winding pattern realized by f is exactly

$$D - \deg(D)[\infty].$$

Therefore the only leftover information in the divisor class is total degree.

$$\boxed{D \sim \deg(D)[\infty]}$$

Why principal divisors have degree zero

Divisor / order language	**Winding / integral language**
$\deg(f) = \sum_p \operatorname{ord}_p(f) = 0$.	$\sum_p \frac{1}{2\pi i} \int_{\gamma_p} \frac{df}{f} = 0$. Total winding on a compact surface must balance.
This follows because principal divisors do not change degree.	This follows because $\frac{df}{f}$ is meromorphic and the sum of its residues is 0.

More explicitly,

$$\operatorname{res}_p\left(\frac{df}{f}\right) = \operatorname{ord}_p(f),$$

so by the residue theorem,

$$\sum_{p \in \mathbf{CP}^1} \operatorname{ord}_p(f) = 0.$$

The theorem in parallel form

Theorem

For every divisor D on $\mathbf{CP}^1$,

$$D \sim \deg(D)[\infty].$$

Proof in divisor language

Write

$$D = \sum_{j=1}^{r} n_j[a_j] + m[\infty].$$

Then

$$\sum_{j=1}^{r} n_j[a_j] \sim \left(\sum_{j=1}^{r} n_j\right)[\infty].$$

Hence

$$D \sim \left(\sum_{j=1}^{r} n_j\right)[\infty] + m[\infty] = \deg(D)[\infty].$$

Proof in winding language

Each factor $z - a_j$ contributes one positive unit of winding at a_j and one negative unit at ∞. Raising to power n_j scales that by n_j. Multiplying all factors adds the winding data. So every prescribed local winding pattern on finite points can be realized by a rational function, and the only necessary balancing term is the negative total winding at ∞. That balancing coefficient is exactly $-\deg(D)$. Hence D is equivalent to concentrating all net winding at ∞, namely $\deg(D)[\infty]$.

Consequences in parallel form

Divisor / order language	**Winding / integral language**
Two divisors D, E are linearly equivalent iff $\deg(D) = \deg(E)$.	Two winding patterns are equivalent iff they have the same total net winding.
$\mathrm{Div}(\mathbf{CP}^1)/\sim \cong \mathbf{Z}$.	A divisor class is determined by the net winding stored at ∞.
Every degree-zero divisor is principal on $\mathbf{CP}^1$.	Any winding pattern with total sum 0 can be realized by a rational function on $\mathbf{CP}^1$.

What is special about $\mathbf{CP}^1$?

Divisor language

On $\mathbf{CP}^1$, every point satisfies

$$[a] - [\infty] = (z - a).$$

So every point is linearly equivalent to ∞.

Winding language

There is a global affine coordinate z on $\mathbf{C} \subset \mathbf{CP}^1$, and the functions $z - a$ let us create one unit of local winding at any chosen finite point, balanced by -1 at ∞. This gives enormous control over local winding data.

On a higher-genus compact Riemann surface, there is no such global coordinate and no analogous family of globally defined functions $z - a$. So arbitrary degree-zero winding data need not come from a global meromorphic function.

Exercises in parallel style

1. Fill in the two-column proof that

$$(z) = [0] - [\infty].$$

2. Fill in the two-column proof that

$$\left(\frac{z-a}{z-b}\right) = [a] - [b].$$

3. Let

$$f(z) = \frac{(z-1)^3(z+2)^5}{(z-i)^2}.$$

Compute (f) in divisor language, then recompute it from loop integrals of $\frac{df}{f}$.

4. Let

$$D = 2[0] - [1] + 4[\infty].$$

Find $\deg(D)$, then explain in both languages why

$$D \sim 5[\infty].$$

5. Explain in both languages why the divisor

$$[0] + [1] + [2] - 3[\infty]$$

is principal.

6. Write a short paragraph explaining the sentence:

"A divisor on $\mathbf{CP}^1$ is local winding data, and degree is the net winding at ∞."

Final slogan

On $\mathbf{CP}^1$, divisor theory and winding-number theory are the same thing:

$$\mathrm{ord}_p(f) = \frac{1}{2\pi i} \int_{\gamma_p} \frac{df}{f}.$$

The function $z - a$ moves one unit of winding from a to ∞. Repeating this for every point appearing in a divisor collapses the divisor to a single coefficient at ∞, namely $\deg(D)$. Thus

$$D \sim \deg(D)[\infty].$$

Two divisors D, E on a Riemann surface X are linearly equivalent if

$$D - E = (f)$$

for some nonzero meromorphic function $f \in \mathcal{M}(X)^\times$.

Using the winding form, this means:

$$D - E = \sum_{p \in X} \left(\frac{1}{2\pi i} \int_{\gamma_p} \frac{df}{f} \right) [p].$$

So $D \sim E$ exactly when the difference between the coefficients of D and E is realized by the local winding numbers of one global meromorphic function.

A divisor D determines a line bundle $\mathcal{O}(D)$. Two divisors D, E are linearly equivalent iff

$$D - E = (g)$$

for some nonzero meromorphic function g. This is equivalent to

$$\mathcal{O}(D) \cong \mathcal{O}(E).$$

Indeed, if $E = D + (g)$, then multiplication by g^{-1} defines an isomorphism

$$\mathcal{O}(D) \xrightarrow{\sim} \mathcal{O}(E).$$

Locally, the coefficient of $[p]$ in $E - D$ is

$$\mathrm{ord}_p(g) = \frac{1}{2\pi i} \int_{\gamma_p} \frac{dg}{g},$$

so the difference between D and E is exactly the local winding data of the meromorphic change of frame g. Thus linearly equivalent divisors define the same holomorphic line bundle.